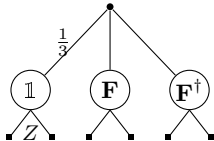
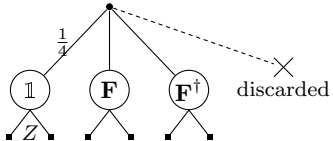


the coin



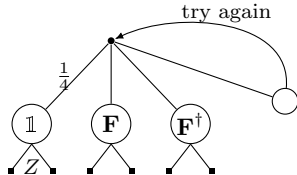
$$p_b = \frac{1}{3}$$

the discard



$$\frac{1}{4} / \frac{3}{4} = \frac{1}{3}$$

the retry



$$\sum_{n \geq 0} \left(\frac{1}{4}\right)^{n+1} = \frac{1}{3}$$